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```
%---  Wing Spar for a Human Powered Aircraft  ---
% MAE 3270 - Fall 2021
% Teammates: Michelle Zhou, George Abbott
%----------------------------------------------

clear; clc; close all
```

Specify loads and constraints

```
Ua = 3;    % [m] Maximum allowable deflection
Xf = 1.4;  % minimum safety factor for strength
Xb = 2.;   % minimum safety factor for local buckling
L = 15;    % [m] wing length
w = 32;    % [N/m] wing loading
g = 9.8;   % [m/s^2] acceleration of gravity
```

Material Properties - enter these from material selected on CES

```
rho = 2851.03;              % REPLACE [kg/m^3] density
E = 72.4*10^9;              % REPLACE [Pa] Young's modulus
nu = 0.34;                  % REPLACE Poisson's ratio
sf = 0.6895*10^9;           % REPLACE [Pa] tensile strength
cost = 4.4;                 % REPLACE [USD/kg] cost
name = 'Aluminum 7068 T6511'; % REPLACE write in material name
```

Section properties - Replace with your own design

example below assumes hollow circular section

```
r0 = 0.03;           % REPLACE [m] outer radius
ri = 0.02;           % REPLACE [m] inner radius

t = r0-ri;              % [m] tube thickness
A = pi*(r0^2-ri^2);     % cross sectional ares
I = pi*(r0^4-ri^4)/4;   % moment of inertia
```

```
r = (ri+r0)/2;    % average radius [m]
Mbuckle =  (0.7*E*r*t^2)/sqrt(1-nu^2);    % REPLACE bending moment
 that will result in local buckling
mass = rho*L*A;
```

Stress and displacement

```
Mmax = (0.5*L^2)*(rho*g*A - w);        % REPLACE max bending moment in
 the wing spar
smax = Mmax*r0/I; % max bending stress


fprintf('Max bending stress = %4.2f MPa \n', smax*1.e-6 )

umax = (rho*g*A - w)*(L^4)/(8*E*I);  % REPLACE max deflection oddurs
 at wing tip
fprintf('Tip Deflection = %4.2f meters \n', umax )
```

```
Max bending stress = 78.59 MPa
Tip Deflection = 2.04 meters
```

Check Safety Factors

```
X = sf/smax;   % REPLACE strength safety factor
XB = Mbuckle/Mmax; % REPLACE buckling safety factor

fprintf('\nStrength Safety Factor = %4.2f, Buckling Safety Factor =
 %4.2f \n', X, XB)
if X > Xf
    fprintf('PASS STRENGTH REQUIREMENT\n')
else
    fprintf('FAIL STRENGTH REQUIREMENT\n')
end
if XB > Xb
    fprintf('PASS BUCKLING REQUIREMENT\n')
else
    fprintf('FAIL BUCKLING REQUIREMENT\n')
end
if umax < Ua
    fprintf('PASS DEFLECTION REQUIREMENT\n')
else
    fprintf('FAIL DEFLECTION REQUIREMENT\n')
end
```

```
Strength Safety Factor = 8.77, Buckling Safety Factor = 100.74
PASS STRENGTH REQUIREMENT
PASS BUCKLING REQUIREMENT
PASS DEFLECTION REQUIREMENT
```

Wrap up

```
fprintf('\nMaterial used is %s \n', name)
```

```
fprintf('Mass = %4.1f kg \n', mass)
fprintf('Material Cost = $ %4.0f  \n', cost*mass)
```

```
Material used is Aluminum 7068 T6511
Mass = 67.2 kg
Material Cost = $  296
```

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